

Pharmacokinetics of Bupropion, a Novel Antidepressant Agent, Following Oral Administration to Healthy Subjects

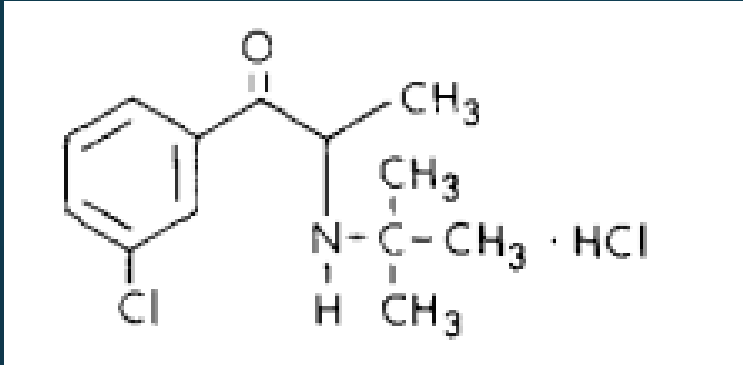
J. W. A. Findlay¹, J. Van Wyck Fleet¹, P. G. Smith¹, R. F. Butz¹, M. L. Hinton¹, M. R. Blum¹, and D. H. Schroeder¹

¹Wellcome Research Laboratories, Research Triangle Park, USA

Introduction

Bupropion is an aminoketone, non-tricyclic antidepressant that affects noradrenaline and dopamine reuptake, with minimal serotonergic effects.

Chemical name of Bupropion : 2-tert-butylamino-3'-chloropropiophenone hydrochloride



Approved for the treatment of major depressive disorder (MDD), and as an aid to smoking cessation (blocks nicotinic acetylcholine receptors).

The authors state that bupropion did not produce significant sympathomimetic effects such as increased heart rate or blood pressure, sympatholytic effects involving reduced sympathetic nervous system activity, anticholinergic effects such as dry mouth, blurred vision, constipation, and urinary retention, or harmful cardiovascular effects on the heart and circulation.

The recent development of a sensitive and specific radioimmunoassay (RIA) made accurate measurement of bupropion in human plasma possible.

In a U.S. outpatient prescribing database based on AHRQ MEPS data, it was estimated to account for 30.3 million prescriptions in 2023, involving about 6.3 million patients; it ranked 17th overall and 3rd among antidepressants in that dataset.¹

Aim: Describe the pharmacokinetics of bupropion in healthy male and female subjects after oral administration of 50, 100, and 200 mg doses of drug in a tablet formation, and to examine urinary excretion of unchanged drug over 24 hours.

Methods

Experiment 1:

Antiserum Production and Assay Procedure

- This is the method for measuring bupropion in plasma.
- Unknown plasma samples and bupropion standards in pooled blank human plasma were analysed in duplicate
- The assay used:
 - ³H-bupropion as the labelled tracer
 - anti-bupropion antiserum diluted in assay buffer (1:5600 dilution)
- Samples were vortex-mixed and incubated overnight at 4°C
- Dextran-coated charcoal used to separate free tracer from antibody-bound tracer
- The bound fraction was measured by scintillation spectrometry
- Standard curves were generated from known bupropion concentrations to determine sample concentrations

Experiment 2:

Determination of Bupropion in Human Urine

- Urine samples were made strongly alkaline (pH: 12-14) with NaOH and Bupropion in urine samples was extracted into hexane
- Bupropion was then back-extracted into dilute HCl, the hexane was evaporated under nitrogen and the acid extract was used for analysis
- Bupropion standards were prepared in pooled blank human urine and processed the same way as samples and analysed in duplicate
- Extracts of standards, controls, and unknown urine samples were incubated with ³H-bupropion and anti-bupropion serum in assay buffer (1:4500 dilution) overnight at 4°C
- Antibody-bound and free radiolabel were separated as in the plasma RIA
- Urinary bupropion concentrations were then calculated from the assay results.

Clinical trials and Results

Clinical studies included two parts:

1.Three-way cross-over study

Subjects: 12 males, 12 females (all healthy)

Age range: 19–43 years old, within 10% of ideal body weight

Each subject received all three oral doses: 50 mg, 100 mg, 200 mg

There was at least a 1-week interval between doses

Subjects fasted from 12 h before until 4 h after dosing

Blood samples were collected: before dosing and at multiple time points up to 24 h

Plasma samples were stored at: −20°C

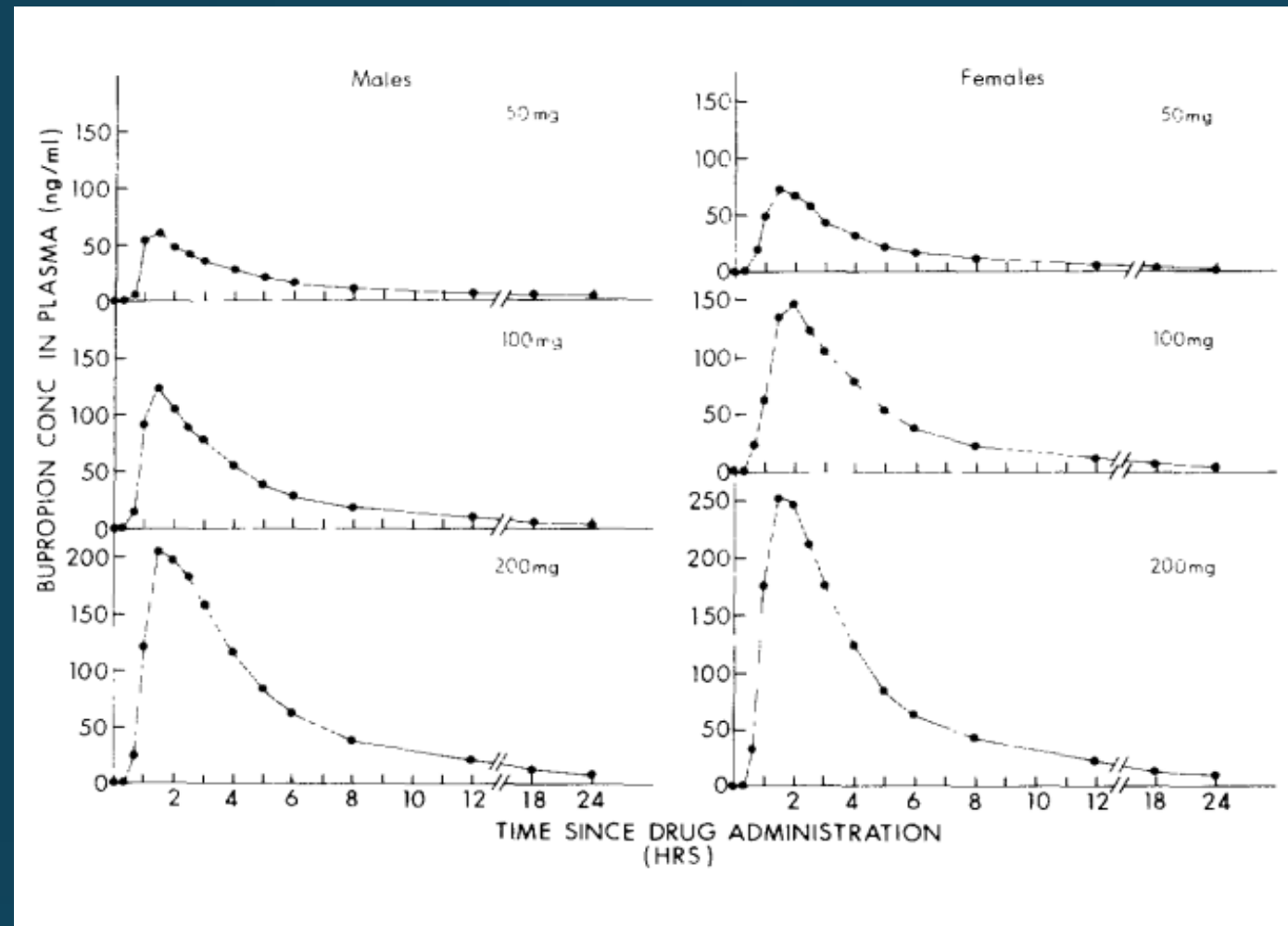


Fig 1: mean plasma concentration–time curves of bupropion in 24 healthy subjects after oral administration

- Mean lag times:** males:0.69 ± 0.03 h females:0.80 ± 0.03 h
- T_{max}**(For all cases) = 3 hrs after drug administration
- At each dose level, higher plasma concentrations were generally attained in females than in males
- C_{max}** increases with the dose in both males and females

2. Urinary excretion study

Subjects: 16 healthy male subjects received 200 mg bupropion hydrochloride as two 100 mg tablets

Urine collected: pre-dose, 0–4 h, 4–6 h, 6–12 h, 12–24 h

excretion volumes recorded for each period

Urine samples were stored at: −20°C

	Amount [µg] of bupropion excreted per time period				Excretion in 24 h	
	0–4 h	4–6 h	6–12 h	12–24 h	[µg]	% dose
Mean	145.3	69.8	77.0	114.7	406.8	0.23
SEM	40.0	23.2	10.4	18.9	56.6	0.03
Mean Excretion Rate [µg/h]	36.3	34.9	12.8	9.6		

Fig 2: Mean 24-hour urinary excretion parameters for bupropion
SEM = how much uncertainty there is around that average

Percentage of dose excreted in 24 h: 0.23%

- So only a very small amount of the administered dose was recovered in urine as unchanged bupropion. According to the mean excretion rate;
- The highest excretion rates occurred in the early periods, especially: 0–4 h, 4–6 h

Parameter	Males			Females		
	43.39 mg Dose ^b	86.78 mg Dose	173.56 mg Dose	43.39 mg Dose	86.78 mg Dose	173.56 mg Dose
AUC [ng·ml ^{−1} ·h]	313±25	621±41	1206±81	386±23 ^c	806±59 ^c	1421±86
AUC / Dose [ng·ml ^{−1} ·h·kg·mg ^{−1}]	484±32	478±24	470±27	513±27	532±30	479±36
C _{max} [ng·ml ^{−1}]	64±5	125±8	223±16	82±6 ^c	158±11 ^c	279±22
C _{max} / Dose [ng·ml ^{−1} ·kg·mg ^{−1}]	100±7	97±5	86±5	110±9	107±8	93±8
V _d /F [l]	538±54	475±20	528±31	374±34 ^c	381±34 ^c	427±31 ^c
V _d /F / Body wt [l·kg ^{−1}]	7.95±0.83	7.01±0.35	7.77±0.44	6.58±0.71	6.67±0.68	7.42±0.59
V _{dss} /F [l]	1297±78	1272±87	1334±92	1020±68 ^c	1113±96	1366±141
V _{dss} /F / Body wt [l·kg ^{−1}]	19.0±1.0	18.6±1.2	19.4±1.0	17.8±1.5	19.5±2.0	23.7±2.5
Cl/F [ml·min ^{−1}]	2458±174	2458±182	2506±147	1957±130 ^c	1906±144 ^c	2128±143 ^c
Cl/F / Body wt [ml·min ^{−1} ·kg ^{−1}]	36.0±2.4	35.9±2.2	36.8±2.3	33.7±2.2	32.6±2.2	37.1±2.9

Fig 3: Some pharmacokinetic parameters for bupropion in normal males and females with and without normalization for individual dose size

Free base equivalent (mg)	Bupropion hydrochloride dose (mg)
43.39	50
86.78	100
173.56	200

AUC = Area under the concentration–time curve

AUC / Dose = AUC corrected for dose size

C_{max} = maximum plasma concentration

C_{max} / Dose = peak concentration corrected for dose size

V_d/F = apparent volume of distribution of the central compartment

V_d/F / Body wt = central distribution volume corrected for body weight

V_{dss}/F = apparent volume of distribution at steady state

V_{dss}/F / Body wt = steady-state distribution volume corrected for body weight

Cl/F = apparent oral clearance

Cl/F / Body wt = clearance corrected for body weight

AUC

- As the dose increases, overall bupropion exposure increases in both sexes.
- At each total dose, females have higher raw AUC values than males.
- Females weighed less on average, so the same fixed dose gave them a higher mg/kg dose.

AUC/Dose

- Much closer values than the raw AUC values.

C_{max}

- As the dose increases, peak plasma concentration increases in both sexes.
- Females have higher raw C_{max} values than males at the same dose (probably related to weight differences)

C_{max}/Dose

- After dose normalisation, relative C_{max} values are constant in both sexes
- There is a slight downward trend in C_{max}/Dose as dose increases.
- Normalised Cmax after the 200 mg dose was about 15% lower than after the 50 mg dose. So peak concentration still rises with dose, but not perfectly proportionally.

V_d/F and V_{dss}/F

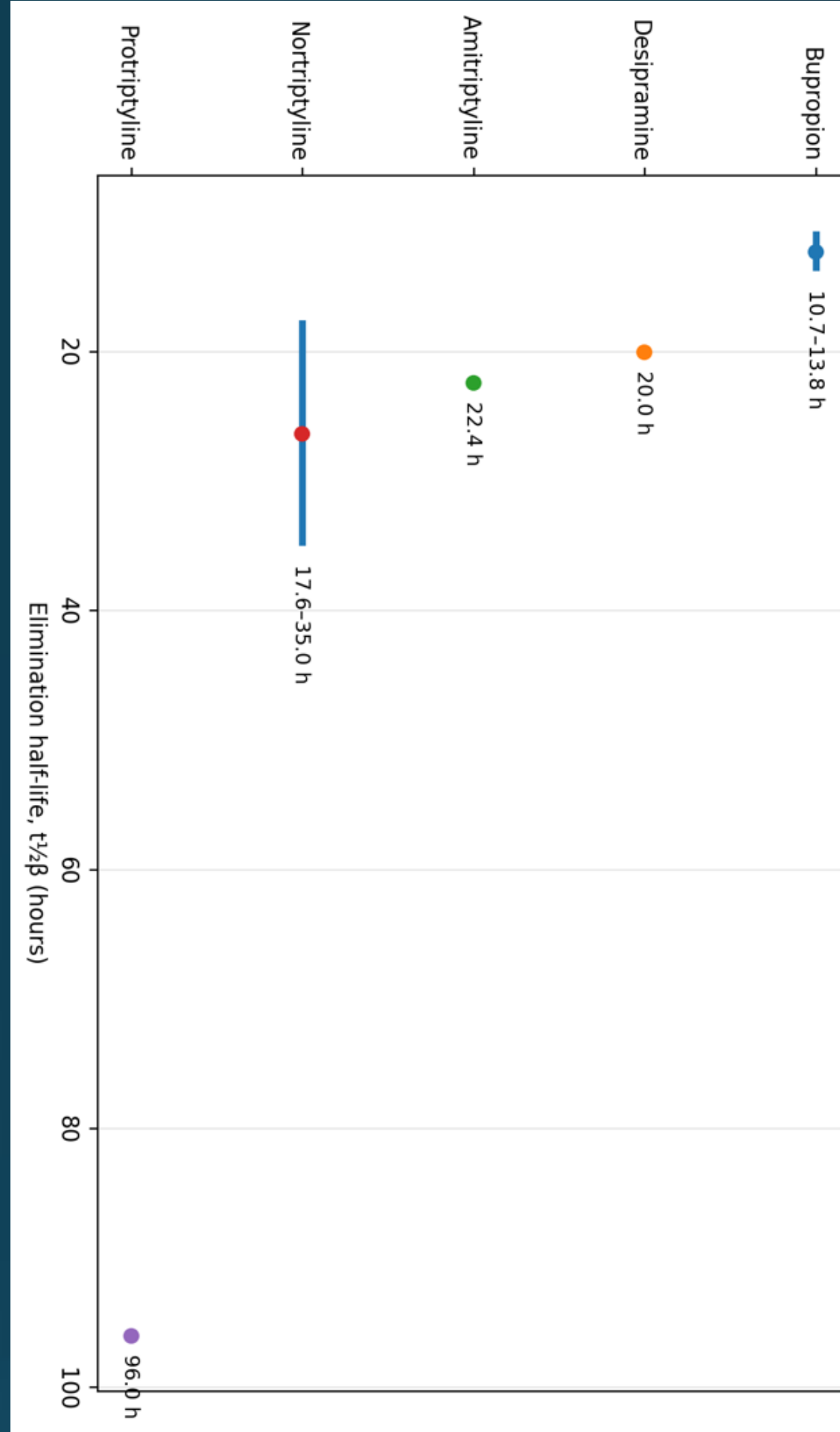
- Before adjusting for weight, females appear to have smaller apparent distribution volumes.
- Once body weight is considered, the apparent sex difference in distribution becomes much less convincing.

Cl/F

- Raw clearance values are lower in females.
- After correcting for body weight, the apparent sex difference in clearance disappears.

Discussion

Compared with the tricyclic antidepressants discussed in the paper, bupropion showed a markedly shorter elimination half-life, suggesting more rapid elimination from plasma.



Limitations of the study:

- The study examined single oral doses only. It does not show what happens during repeated dosing or at steady state, which is more relevant clinically for antidepressants
- Only unchanged bupropion was measured in urine, but it did not fully quantify all metabolites, even though the discussion suggests extensive metabolism.

Future directions:

- Investigate whether the pharmacokinetics are the same in the actual target population (patients with depression)
- Study pharmacokinetics in people with hepatic or renal impairment, since the paper suggests metabolism is mainly hepatic
- Since antidepressants are often used with other medicines, interaction studies would be clinically useful